Improper Integrals

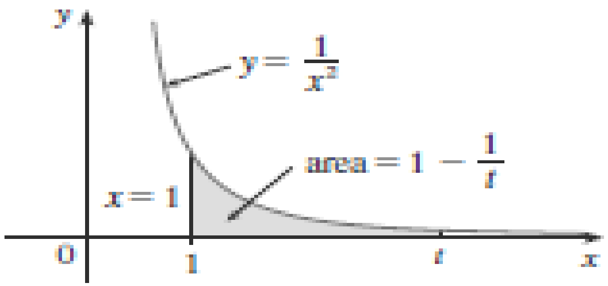
In this section we extend the concept of a definite integral to the case where the interval is infinite and also to the case where f has an infinite discontinuity in $[a, b]$. In either case the integral is called an *improper* integral.

Type I: Infinite Intervals

Consider the infinite region S that lies under the curve $y = 1/x^2$, above the x-axis, and to the right of the line $x = 1$. You might think that, since S is infinite in extent, its area must be infinite, but let's take a closer look. The area of the part of S that lies to the left of the line $x = t$ (shaded in Figure) is

$$A(t) = \int_1^t \frac{1}{x^2}\,dx = -\frac{1}{x}\Big]_1^t = 1 - \frac{1}{t}$$

Notice that $A(t) < 1$ no matter how large t is chosen.



We also observe that

$$\lim_{t\to\infty} A(t) = \lim_{t\to\infty}\left(1 - \frac{1}{t}\right) = 1$$

The area of the shaded region approaches 1 as $t \to \infty$, so we say that the area of the infinite region S is equal to 1 and we write

$$\int_1^\infty \frac{1}{x^2}\,dx = \lim_{t\to\infty}\int_1^t \frac{1}{x^2}\,dx = 1$$

DEFINITION 1

Improper integrals of type I

If f is continuous on $[a, \infty)$, we define the improper integral of f over $[a, \infty)$ as a limit of proper integrals:

$$\int_a^\infty f(x)\,dx = \lim_{R\to\infty} \int_a^R f(x)\,dx.$$

Similarly, if f is continuous on $(-\infty, b]$, then we define

$$\int_{-\infty}^b f(x)\,dx = \lim_{R\to-\infty} \int_R^b f(x)\,dx.$$

In either case, if the limit exists (is a finite number), we say that the improper integral **converges**; if the limit does not exist, we say that the improper integral **diverges**. If the limit is ∞ (or $-\infty$), we say the improper integral **diverges to infinity** (or **diverges to negative infinity**).

The integral $\int_{-\infty}^\infty f(x)\,dx$ is, for f continuous on the real line, improper of type I at both endpoints. We break it into two separate integrals:

$$\int_{-\infty}^\infty f(x)\,dx = \int_{-\infty}^0 f(x)\,dx + \int_0^\infty f(x)\,dx.$$

The integral on the left converges if and only if *both* integrals on the right converge.

EX.

Find the area of the region A lying under the curve $y = 1/x^2$ and above the x-axis to the right of $x = 1$. (See Figure (a).)

We would like to calculate the area with an integral

$$A = \int_1^\infty \frac{dx}{x^2},$$

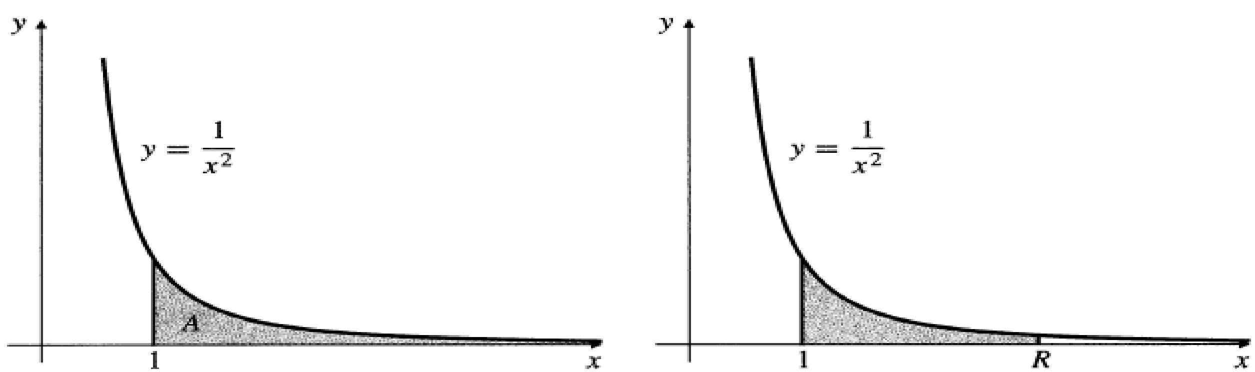
$$A = \int_1^\infty \frac{dx}{x^2} = \lim_{R\to\infty} \int_1^R \frac{dx}{x^2} = \lim_{R\to\infty} \left(-\frac{1}{x}\right)\Bigg|_1^R$$
$$= \lim_{R\to\infty} \left(-\frac{1}{R} + 1\right) = 1$$

Since the limit exists (is finite), we say that the improper integral *converges*. The region has finite area $A = 1$ square unit.

Figure

(a) $A = \int_1^\infty \frac{1}{x^2}\,dx$

(b) $A = \lim_{R\to\infty} \int_1^R \frac{1}{x^2}\,dx$



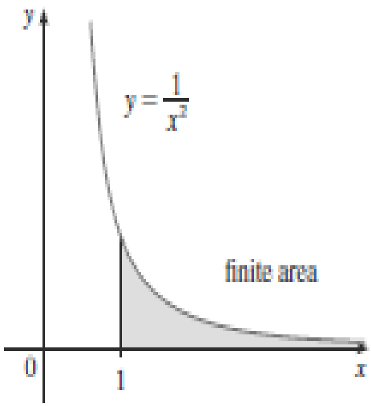
(a) (b)

EXAMPLE 1 Determine whether the integral $\int_1^\infty (1/x)\,dx$ is convergent or divergent.

SOLUTION we have

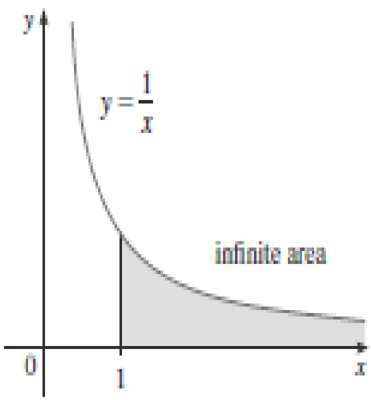
$$\int_1^\infty \frac{1}{x}\,dx = \lim_{t\to\infty}\int_1^t \frac{1}{x}\,dx = \lim_{t\to\infty} \ln|x|\Big]_1^t$$

$$= \lim_{t\to\infty}(\ln t - \ln 1) = \lim_{t\to\infty}\ln t = \infty$$



The limit does not exist as a finite number and so the improper integral $\int_1^\infty (1/x)\,dx$ is divergent.

Let's compare the result of Example 1 with the example given at the beginning of this section:



$$\int_1^\infty \frac{1}{x^2}\,dx \text{ converges} \qquad \int_1^\infty \frac{1}{x}\,dx \text{ diverges}$$

Geometrically, this says that although the curves $y = 1/x^2$ and $y = 1/x$ look very similar for $x > 0$, the region under $y = 1/x^2$ to the right of $x = 1$ (the shaded region in Figure above) has finite area whereas the corresponding region under $y = 1/x$ (in Figure) has infinite area. Note that both $1/x^2$ and $1/x$ approach 0 as $x \to \infty$ but $1/x^2$ approaches 0 faster than $1/x$. The values of $1/x$ don't decrease fast enough for its integral to have a finite value.

EXAMPLE 2 Evaluate $\int_{-\infty}^{0} xe^x\,dx$.

SOLUTION , we have

$$\int_{-\infty}^{0} xe^x\,dx = \lim_{t\to-\infty}\int_{t}^{0} xe^x\,dx$$

We integrate by parts with $u = x$, $dv = e^x\,dx$ so that $du = dx$, $v = e^x$:

$$\begin{aligned}\int_{t}^{0} xe^x\,dx &= xe^x\Big]_t^0 - \int_{t}^{0} e^x\,dx \\ &= -te^t - 1 + e^t\end{aligned}$$

We know that $e^t \to 0$ as $t \to -\infty$, and by l'Hospital's Rule we have

$$\begin{aligned}\lim_{t\to-\infty} te^t &= \lim_{t\to-\infty}\frac{t}{e^{-t}} = \lim_{t\to-\infty}\frac{1}{-e^{-t}} \\ &= \lim_{t\to-\infty}(-e^t) = 0\end{aligned}$$

Therefore

$$\begin{aligned}\int_{-\infty}^{0} xe^x\,dx &= \lim_{t\to-\infty}(-te^t - 1 + e^t) \\ &= -0 - 1 + 0 = -1\end{aligned}$$

EXAMPLE 3 Evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx$.

SOLUTION It's convenient to choose $a = 0$

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx = \int_{-\infty}^{0} \frac{1}{1+x^2}\,dx + \int_{0}^{\infty} \frac{1}{1+x^2}\,dx$$

We must now evaluate the integrals on the right side separately:

$$\int_{0}^{\infty} \frac{1}{1+x^2}\,dx = \lim_{t\to\infty} \int_{0}^{t} \frac{dx}{1+x^2} = \lim_{t\to\infty} \tan^{-1}x\Big]_0^t$$

$$= \lim_{t\to\infty} (\tan^{-1}t - \tan^{-1}0) = \lim_{t\to\infty} \tan^{-1}t = \frac{\pi}{2}$$

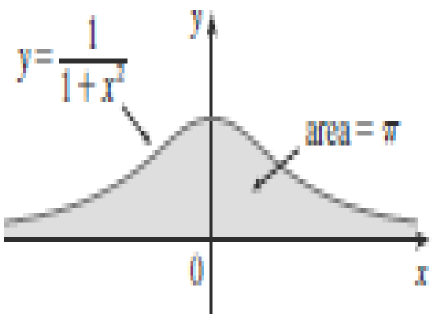
$$\int_{-\infty}^{0} \frac{1}{1+x^2}\,dx = \lim_{t\to-\infty} \int_{t}^{0} \frac{dx}{1+x^2} = \lim_{t\to-\infty} \tan^{-1}x\Big]_t^0$$

$$= \lim_{t\to-\infty} (\tan^{-1}0 - \tan^{-1}t)$$

$$= 0 - \left(-\frac{\pi}{2}\right) = \frac{\pi}{2}$$

Since both of these integrals are convergent, the given integral is convergent and

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx = \frac{\pi}{2} + \frac{\pi}{2} = \pi$$



FIGURE

Since $1/(1 + x^2) > 0$, the given improper integral can be interpreted as the area of the infinite region that lies under the curve $y = 1/(1 + x^2)$ and above the x-axis (see Figure

OR

Evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{1}{1+x^2}\, dx.$

By the (even) symmetry of the integrand

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \int_{-\infty}^{0} \frac{dx}{1+x^2} + \int_{0}^{\infty} \frac{dx}{1+x^2}$$

$$= 2 \lim_{R\to\infty} \int_{0}^{R} \frac{dx}{1+x^2}$$

$$= 2 \lim_{R\to\infty} \tan^{-1} R = 2\left(\frac{\pi}{2}\right) = \pi.$$

Definition of an Improper Integral of Type 1

(a) If $\int_a^t f(x)\,dx$ exists for every number $t \geqslant a$, then

$$\int_a^\infty f(x)\,dx = \lim_{t \to \infty} \int_a^t f(x)\,dx$$

provided this limit exists (as a finite number).

(b) If $\int_t^b f(x)\,dx$ exists for every number $t \leqslant b$, then

$$\int_{-\infty}^b f(x)\,dx = \lim_{t \to -\infty} \int_t^b f(x)\,dx$$

provided this limit exists (as a finite number).

The improper integrals $\int_a^\infty f(x)\,dx$ and $\int_{-\infty}^b f(x)\,dx$ are called **convergent** if the corresponding limit exists and **divergent** if the limit does not exist.

(c) If both $\int_a^\infty f(x)\,dx$ and $\int_{-\infty}^a f(x)\,dx$ are convergent, then we define

$$\int_{-\infty}^\infty f(x)\,dx = \int_{-\infty}^a f(x)\,dx + \int_a^\infty f(x)\,dx$$

EXAMPLE 4 For what values of p is the integral

$$\int_1^\infty \frac{1}{x^p}\,dx$$

convergent?

SOLUTION We know from Example 1 that if $p = 1$, then the integral is divergent, so let's assume that $p \neq 1$. Then

$$\begin{aligned}
\int_1^\infty \frac{1}{x^p}\,dx &= \lim_{t\to\infty} \int_1^t x^{-p}\,dx \\
&= \lim_{t\to\infty} \frac{x^{-p+1}}{-p+1}\bigg]_{x=1}^{x=t} \\
&= \lim_{t\to\infty} \frac{1}{1-p}\left[\frac{1}{t^{p-1}} - 1\right]
\end{aligned}$$

If $p > 1$, then $p - 1 > 0$, so as $t \to \infty$, $t^{p-1} \to \infty$ and $1/t^{p-1} \to 0$. Therefore

$$\int_1^\infty \frac{1}{x^p}\,dx = \frac{1}{p-1} \qquad \text{if } p > 1$$

and so the integral converges. But if $p < 1$, then $p - 1 < 0$ and so

$$\frac{1}{t^{p-1}} = t^{1-p} \to \infty \qquad \text{as } t \to \infty$$

and the integral diverges.

We summarize the result of Example 4 for future reference:

$\boxed{2}$	$\int_1^\infty \frac{1}{x^p}\,dx$ is convergent if $p > 1$ and divergent if $p \leq 1$.

Type 2: Discontinuous Integrands

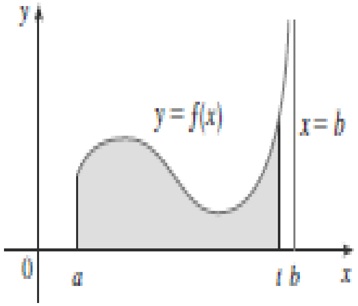
Suppose that f is a positive continuous function defined on a finite interval $[a, b)$ but has a vertical asymptote at b. Let S be the unbounded region under the graph of f and above the x-axis between a and b. (For Type 1 integrals, the regions extended indefinitely in a horizontal direction. Here the region is infinite in a vertical direction.) The area of the part of S between a and t (the shaded region in Figure *) is

$$A(t) = \int_a^t f(x)\,dx$$

If it happens that $A(t)$ approaches a definite number A as $t \to b^-$, then we say that the area of the region S is A and we write

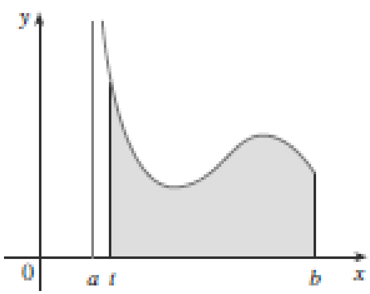
$$\int_a^b f(x)\,dx = \lim_{t \to b^-} \int_a^t f(x)\,dx$$

We use this equation to define an improper integral of Type 2 even when f is not a positive function, no matter what type of discontinuity f has at b.

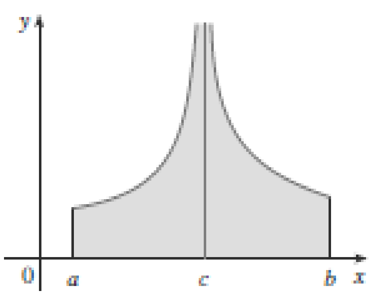


FIGURE*

: where $f(x) \geq$ (
and f has vertical asymptotes at a and c, respectively.



FIGURE



FIGURE

3 Definition of an Improper Integral of Type 2

(a) If f is continuous on $[a, b)$ and is discontinuous at b, then

$$\int_a^b f(x)\,dx = \lim_{t \to b^-} \int_a^t f(x)\,dx$$

if this limit exists (as a finite number).

(b) If f is continuous on $(a, b]$ and is discontinuous at a, then

$$\int_a^b f(x)\,dx = \lim_{t \to a^+} \int_t^b f(x)\,dx$$

if this limit exists (as a finite number).

The improper integral $\int_a^b f(x)\,dx$ is called **convergent** if the corresponding limit exists and **divergent** if the limit does not exist.

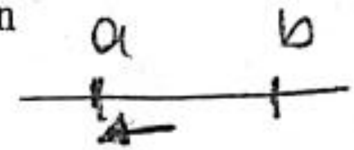
(c) If f has a discontinuity at c, where $a < c < b$, and both $\int_a^c f(x)\,dx$ and $\int_c^b f(x)\,dx$ are convergent, then we define

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

DEFINITION Integrals of functions that become infinite at a point within the interval of integration are **improper integrals of Type II.**

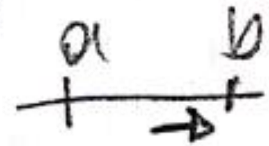
1. If $f(x)$ is continuous on $(a, b]$ and discontinuous at a, then

$$\int_a^b f(x)\,dx = \lim_{c \to a^+} \int_c^b f(x)\,dx.$$



2. If $f(x)$ is continuous on $[a, b)$ and discontinuous at b, then

$$\int_a^b f(x)\,dx = \lim_{c \to b^-} \int_a^c f(x)\,dx.$$



3. If $f(x)$ is discontinuous at c, where $a < c < b$, and continuous on $[a, c) \cup (c, b]$, then

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.$$

In each case, if the limit is finite we say the improper integral **converges** and that the limit is the **value** of the improper integral. If the limit does not exist, the integral **diverges**.

Investigate the convergence of

$$\int_0^1 \frac{1}{1-x}\,dx.$$

$0 \to 1.$

Solution The integrand $f(x) = 1/(1-x)$ is continuous on $[0, 1)$ but is discontinuous at $x = 1$ and becomes infinite as $x \to 1^-$ (. We evaluate the integral as

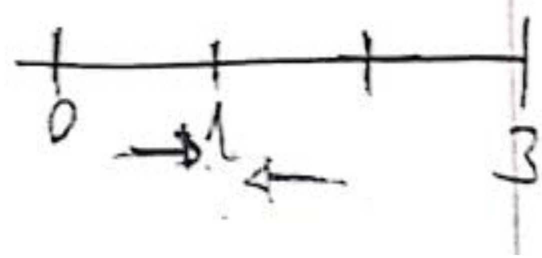
$$\lim_{b\to 1^-}\int_0^b \frac{1}{1-x}\,dx = \lim_{b\to 1^-}\Big[-\ln|1-x|\Big]_0^b$$

$$= \lim_{b\to 1^-}[-\ln(1-b) + 0] = \infty.$$

The limit is infinite, so the integral diverges. ✓

Evaluate

$$\int_0^3 \frac{dx}{(x-1)^{2/3}}.$$

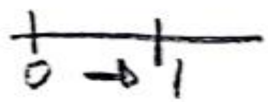


Solution The integrand has a vertical asymptote at $x = 1$ and is continuous on $[0, 1)$ and $(1, 3]$ (Figure). Thus, by Part 3 of the definition above,

$$\int_0^3 \frac{dx}{(x-1)^{2/3}} = \int_0^1 \frac{dx}{(x-1)^{2/3}} + \int_1^3 \frac{dx}{(x-1)^{2/3}}.$$

Next, we evaluate each improper integral on the right-hand side of this equation.

$$\int_0^1 \frac{dx}{(x-1)^{2/3}} = \lim_{b\to 1^-} \int_0^b \frac{dx}{(x-1)^{2/3}}$$



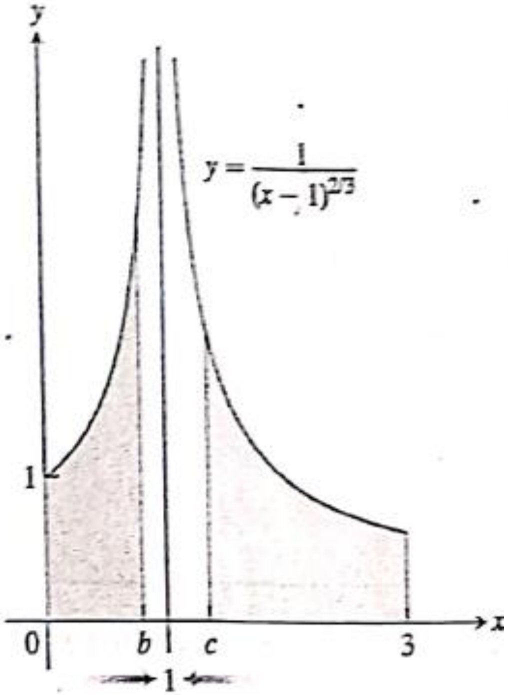
$$= \lim_{b\to 1^-} 3(x-1)^{1/3}\Big]_0^b$$

$$= \lim_{b\to 1^-} [3(b-1)^{1/3} + 3] = 3$$

$$\int_1^3 \frac{dx}{(x-1)^{2/3}} = \lim_{c\to 1^+} \int_c^3 \frac{dx}{(x-1)^{2/3}}$$

$$= \lim_{c\to 1^+} 3(x-1)^{1/3}\Big]_c^3$$

$$= \lim_{c\to 1^+} \left[3(3-1)^{1/3} - 3(c-1)^{1/3}\right] = 3\sqrt[3]{2}$$



FIGURE

$$\int_0^3 \frac{dx}{(x-1)^{2/3}} = 3 + 3\sqrt[3]{2}.$$

$$\begin{aligned}\int_0^a x^{-p}\,dx &= \lim_{c\to 0+}\int_c^a x^{-p}\,dx\\ &= \lim_{c\to 0+}\left.\frac{x^{-p+1}}{-p+1}\right|_c^a\\ &= \lim_{c\to 0+}\frac{a^{1-p}-c^{1-p}}{1-p} = \frac{a^{1-p}}{1-p}\end{aligned}$$

because $1 - p > 0$. If $p > 1$, then

$$\begin{aligned}\int_0^a x^{-p}\,dx &= \lim_{c\to 0+}\int_c^a x^{-p}\,dx\\ &= \lim_{c\to 0+}\left.\frac{x^{-p+1}}{-p+1}\right|_c^a\\ &= \lim_{c\to 0+}\frac{c^{-(p-1)}-a^{-(p-1)}}{p-1} = \infty.\end{aligned}$$

****

p-integrals

If $0 < a < \infty$, then

(a) $$\int_a^\infty x^{-p}\,dx \quad \begin{cases} \text{converges to } \dfrac{a^{1-p}}{p-1} & \text{if } p > 1 \\ \text{diverges to } \infty & \text{if } p \le 1 \end{cases}$$

(b) $$\int_0^a x^{-p}\,dx \quad \begin{cases} \text{converges to } \dfrac{a^{1-p}}{1-p} & \text{if } p < 1 \\ \text{diverges to } \infty & \text{if } p \ge 1. \end{cases}$$

EX. Find the area of the region S lying under $y = 1/\sqrt{x}$, above the x-axis, between $x = 0$ and $x = 1$.

$$A = \int_0^1 \frac{1}{\sqrt{x}}\,dx,$$

$$A = \lim_{c\to 0+} \int_c^1 x^{-1/2}\,dx = \lim_{c\to 0+} 2x^{1/2}\Big|_c^1 = \lim_{c\to 0+} (2 - 2\sqrt{c}) = 2.$$

This integral converges, and S has a finite area of 2 square units.

EX. Evaluate each of the following integrals or show that it diverges:

(a) $\displaystyle\int_0^1 \frac{1}{x}\,dx,$ (b) $\displaystyle\int_0^2 \frac{1}{\sqrt{2x - x^2}}\,dx,$ and (c) $\displaystyle\int_0^1 \ln x\,dx.$

Solution

(a) $$\int_0^1 \frac{1}{x}\,dx = \lim_{c\to 0+} \int_c^1 \frac{1}{x}\,dx = \lim_{c\to 0+} (\ln 1 - \ln c) = \infty.$$

This integral diverges to infinity.

(b) $$\int_0^2 \frac{1}{\sqrt{2x-x^2}}\,dx = \int_0^2 \frac{1}{\sqrt{1-(x-1)^2}}\,dx \qquad \text{Let } u = x-1,\ du = dx$$

$$= \int_{-1}^1 \frac{1}{\sqrt{1-u^2}}\,du$$

$$= 2\int_0^1 \frac{1}{\sqrt{1-u^2}}\,du \qquad \text{(by symmetry)}$$

$$= 2 \lim_{c\to 1-} \int_0^c \frac{1}{\sqrt{1-u^2}}\,du$$

$$= 2 \lim_{c\to 1-} \sin^{-1} u \Big|_0^c = 2 \lim_{c\to 1-} \sin^{-1} c = \pi.$$

This integral converges to π. Observe how a change of variable can be made even before an improper integral is expressed as a limit of proper integrals.

(c) $$\int_0^1 \ln x\,dx = \lim_{c\to 0+} \int_c^1 \ln x\,dx$$

$$= \lim_{c\to 0+} (x\ln x - x)\Big|_c^1$$

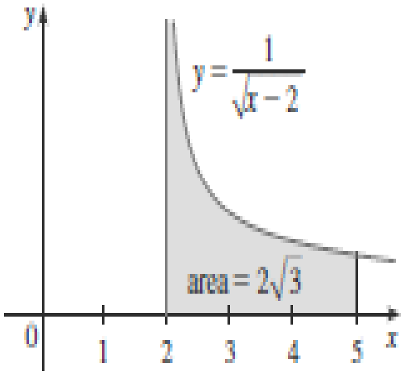
$$= \lim_{c\to 0+} (0 - 1 - c\ln c + c)$$

$$= -1 + 0 - \lim_{c\to 0+} \frac{\ln c}{1/c} \qquad \left[\frac{-\infty}{\infty}\right]$$

$$= -1 - \lim_{c\to 0+} \frac{1/c}{-(1/c^2)} \qquad \text{(by l'Hôpital's Rule)}$$

$$= -1 - \lim_{c\to 0+} (-c) = -1 + 0 = -1.$$

The integral converges to -1.



FIGURE

EXAMPLE Find $\int_2^5 \frac{1}{\sqrt{x-2}}\,dx$.

SOLUTION We note first that the given integral is improper because $f(x) = 1/\sqrt{x-2}$ has the vertical asymptote $x = 2$. Since the infinite discontinuity occurs at the left endpoint of $[2, 5]$,

$$\begin{aligned}\int_2^5 \frac{dx}{\sqrt{x-2}} &= \lim_{t\to 2^+} \int_t^5 \frac{dx}{\sqrt{x-2}} \\ &= \lim_{t\to 2^+} 2\sqrt{x-2}\Big]_t^5 \\ &= \lim_{t\to 2^+} 2\left(\sqrt{3} - \sqrt{t-2}\right) \\ &= 2\sqrt{3}\end{aligned}$$

Thus, the given improper integral is convergent and, since the integrand is positive, we can interpret the value of the integral as the area of the shaded region in Figure

EXAMPLE Determine whether $\int_0^{\pi/2} \sec x\,dx$ converges or diverges.

SOLUTION Note that the given integral is improper because $\lim_{x\to(\pi/2)^-} \sec x = \infty$.

$$\begin{aligned}\int_0^{\pi/2} \sec x\,dx &= \lim_{t\to(\pi/2)^-} \int_0^t \sec x\,dx \\ &= \lim_{t\to(\pi/2)^-} \ln|\sec x + \tan x|\Big]_0^t \\ &= \lim_{t\to(\pi/2)^-} [\ln(\sec t + \tan t) - \ln 1] \\ &= \infty\end{aligned}$$

because $\sec t \to \infty$ and $\tan t \to \infty$ as $t \to (\pi/2)^-$. Thus, the given improper integral is divergent.

EXAMPLE Evaluate $\int_0^3 \frac{dx}{x-1}$ if possible.

SOLUTION Observe that the line $x = 1$ is a vertical asymptote of the integrand. Since it occurs in the middle of the interval $[0, 3]$, we must use part (c) of Definition 3 with $c = 1$:

$$\int_0^3 \frac{dx}{x-1} = \int_0^1 \frac{dx}{x-1} + \int_1^3 \frac{dx}{x-1}$$

where

$$\int_0^1 \frac{dx}{x-1} = \lim_{t \to 1^-} \int_0^t \frac{dx}{x-1} = \lim_{t \to 1^-} \ln|x-1| \Big]_0^t$$

$$= \lim_{t \to 1^-} (\ln|t-1| - \ln|-1|)$$

$$= \lim_{t \to 1^-} \ln(1-t) = -\infty$$

because $1 - t \to 0^+$ as $t \to 1^-$. Thus, $\int_0^1 dx/(x-1)$ is divergent. This implies that $\int_0^3 dx/(x-1)$ is divergent. [We do not need to evaluate $\int_1^3 dx/(x-1)$.]

WARNING • If we had not noticed the asymptote $x = 1$ in Example this and had instead confused the integral with an ordinary integral, then we might have made the following erroneous calculation:

$$\int_0^3 \frac{dx}{x-1} = \ln|x-1| \Big]_0^3 = \ln 2 - \ln 1 = \ln 2$$

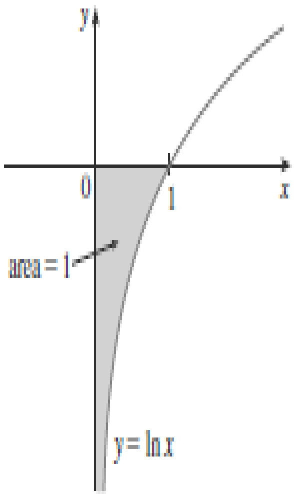
This is wrong because the integral is improper and must be calculated in terms of limits.

From now on, whenever you meet the symbol $\int_a^b f(x)\,dx$ you must decide, by looking at the function f on $[a, b]$, whether it is an ordinary definite integral or an improper integral.

EXAMPLE Evaluate $\int_0^1 \ln x\,dx$.

SOLUTION We know that the function $f(x) = \ln x$ has a vertical asymptote at 0 since $\lim_{x\to 0^+} \ln x = -\infty$. Thus, the given integral is improper and we have

$$\int_0^1 \ln x\,dx = \lim_{t\to 0^+} \int_t^1 \ln x\,dx$$



FIGURE

Now we integrate by parts with $u = \ln x$, $dv = dx$, $du = dx/x$, and $v = x$:

$$\begin{aligned}\int_t^1 \ln x\,dx &= x \ln x\Big]_t^1 - \int_t^1 dx \\ &= 1 \ln 1 - t \ln t - (1 - t) \\ &= -t \ln t - 1 + t\end{aligned}$$

To find the limit of the first term we use l'Hospital's Rule:

$$\begin{aligned}\lim_{t\to 0^+} t \ln t &= \lim_{t\to 0^+} \frac{\ln t}{1/t} \\ &= \lim_{t\to 0^+} \frac{1/t}{-1/t^2} \\ &= \lim_{t\to 0^+} (-t) = 0\end{aligned}$$

Therefore
$$\begin{aligned}\int_0^1 \ln x\,dx &= \lim_{t\to 0^+} (-t \ln t - 1 + t) \\ &= -0 - 1 + 0 = -1\end{aligned}$$

Figure shows the geometric interpretation of this result. The area of the shaded region above $y = \ln x$ and below the x-axis is 1.

EX. Evaluate $\int_0^2 f(x)\,dx$, where $f(x) = \begin{cases} 1/\sqrt{x} & \text{if } 0 < x \le 1 \\ x - 1 & \text{if } 1 < x \le 2. \end{cases}$

$$\int_0^2 f(x)\,dx = \int_0^1 \frac{dx}{\sqrt{x}} + \int_1^2 (x-1)\,dx$$

$$= \lim_{c \to 0+} \int_c^1 \frac{dx}{\sqrt{x}} + \left(\frac{x^2}{2} - x\right)\Bigg|_1^2 = 2 + \left(2 - 2 - \frac{1}{2} + 1\right) = \frac{5}{2};$$

the first integral on the right is improper but convergent

the second is proper.

The Integral $\int_1^{\infty} \frac{dx}{x^p}$

The function $y = 1/x$ is the boundary between the convergent and divergent improper integrals with integrands of the form $y = 1/x^p$. As the next example shows, the improper integral converges if $p > 1$ and diverges if $p \leq 1$.

EXAMPLE For what values of p does the integral $\int_1^{\infty} dx/x^p$ converge? When the integral does converge, what is its value?

Solution If $p \neq 1$,

$$\int_1^b \frac{dx}{x^p} = \frac{x^{-p+1}}{-p+1}\bigg]_1^b = \frac{1}{1-p}(b^{-p+1} - 1) = \frac{1}{1-p}\left(\frac{1}{b^{p-1}} - 1\right).$$

Thus,

$$\int_1^{\infty} \frac{dx}{x^p} = \lim_{b\to\infty} \int_1^b \frac{dx}{x^p}$$

$$= \lim_{b\to\infty}\left[\frac{1}{1-p}\left(\frac{1}{b^{p-1}} - 1\right)\right] = \begin{cases} \frac{1}{p-1}, & p > 1 \\ \infty, & p < 1 \end{cases}$$

because

$$\lim_{b\to\infty} \frac{1}{b^{p-1}} = \begin{cases} 0, & p > 1 \\ \infty, & p < 1. \end{cases}$$

Therefore, the integral converges to the value $1/(p-1)$ if $p > 1$ and it diverges if $p < 1$.

If $p = 1$, the integral also diverges:

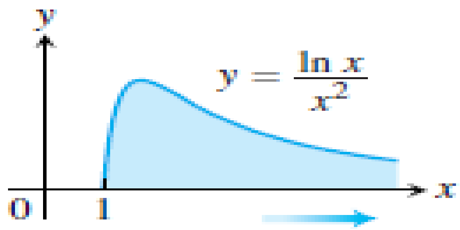
$$\begin{aligned}
\int_1^\infty \frac{dx}{x^p} = \int_1^\infty \frac{dx}{x} \\
&= \lim_{b\to\infty} \int_1^b \frac{dx}{x} \\
&= \lim_{b\to\infty} \ln x \Big]_1^b \\
&= \lim_{b\to\infty} (\ln b - \ln 1) = \infty .
\end{aligned}$$

INFINITE LIMITS OF INTEGRATION: TYPE I

REVIEW

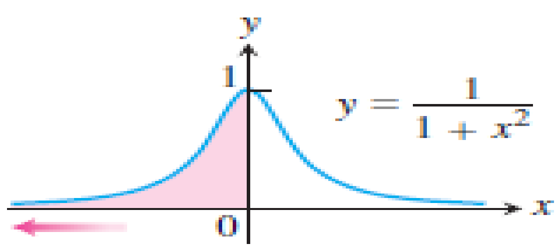
1. Upper limit

$$\int_1^{\infty} \frac{\ln x}{x^2}\,dx = \lim_{b\to\infty} \int_1^{b} \frac{\ln x}{x^2}\,dx$$



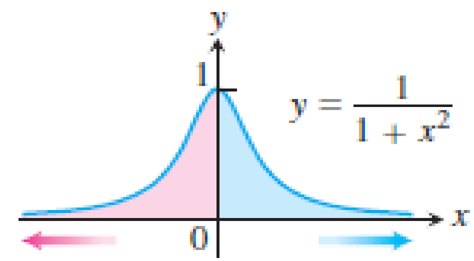
2. Lower limit

$$\int_{-\infty}^{0} \frac{dx}{1+x^2} = \lim_{a\to-\infty} \int_a^{0} \frac{dx}{1+x^2}$$



3. Both limits

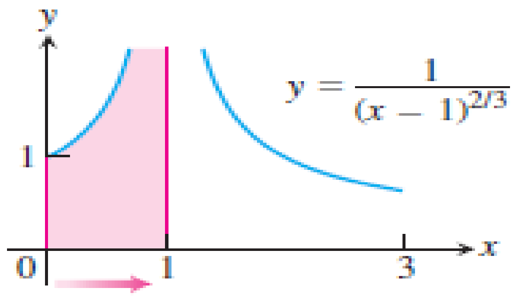
$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \lim_{b\to-\infty} \int_b^{0} \frac{dx}{1+x^2} + \lim_{c\to\infty} \int_0^{c} \frac{dx}{1+x^2}$$



INTEGRAND BECOMES INFINITE: TYPE II

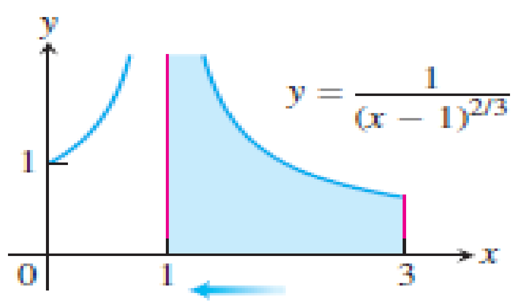
4. Upper endpoint

$$\int_0^1 \frac{dx}{(x-1)^{2/3}} = \lim_{b \to 1^-} \int_0^b \frac{dx}{(x-1)^{2/3}}$$



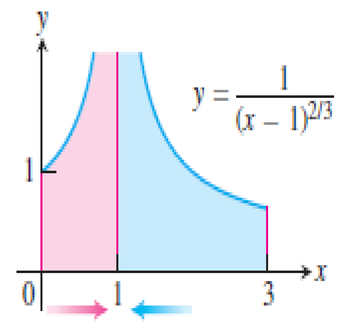
5. Lower endpoint

$$\int_1^3 \frac{dx}{(x-1)^{2/3}} = \lim_{d \to 1^+} \int_d^3 \frac{dx}{(x-1)^{2/3}}$$



6. Interior point

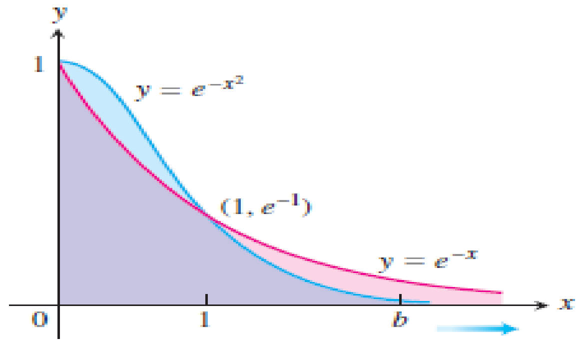
$$\int_0^3 \frac{dx}{(x-1)^{2/3}} = \int_0^1 \frac{dx}{(x-1)^{2/3}} + \int_1^3 \frac{dx}{(x-1)^{2/3}}$$



Tests for Convergence and Divergence

When we cannot evaluate an improper integral directly, we try to determine whether it converges or diverges. If the integral diverges, that's the end of the story. If it converges, we can use numerical methods to approximate its value. The principal tests for convergence or divergence are the Direct Comparison Test and the Limit Comparison Test.

EXAMPLE Does the integral $\int_1^\infty e^{-x^2}\,dx$ converge?



The graph of e^{-x^2} lies below the graph of e^{-x} for $x > 1$

Solution By definition,

$$\int_1^\infty e^{-x^2}\,dx = \lim_{b\to\infty}\int_1^b e^{-x^2}\,dx.$$

We cannot evaluate this integral directly because it is nonelementary. But we *can* show that its limit as $b \to \infty$ is finite. We know that $\int_1^b e^{-x^2}\,dx$ is an increasing function of b. Therefore either it becomes infinite as $b \to \infty$ or it has a finite limit as $b \to \infty$. It does not become infinite: For every value of $x \geq 1$, we have $e^{-x^2} \leq e^{-x}$ so that

$$\int_1^b e^{-x^2}\,dx \leq \int_1^b e^{-x}\,dx = -e^{-b} + e^{-1} < e^{-1} \approx 0.36788.$$

Hence,

$$\int_1^\infty e^{-x^2}\,dx = \lim_{b\to\infty}\int_1^b e^{-x^2}\,dx$$

converges to some definite finite value. We do not know exactly what the value is except that it is something positive and less than 0.37.

THEOREM —Direct Comparison Test Let f and g be continuous on $[a, \infty)$ with $0 \le f(x) \le g(x)$ for all $x \ge a$. Then

1. $\displaystyle\int_a^\infty f(x)\,dx$ converges if $\displaystyle\int_a^\infty g(x)\,dx$ converges.

2. $\displaystyle\int_a^\infty g(x)\,dx$ diverges if $\displaystyle\int_a^\infty f(x)\,dx$ diverges.

EXAMPLE These examples illustrate how we use Theorem .

(a) $\displaystyle\int_1^{\infty} \frac{\sin^2 x}{x^2}\,dx$ converges because

$$0 \le \frac{\sin^2 x}{x^2} \le \frac{1}{x^2} \quad \text{on} \quad [1, \infty) \quad \text{and} \quad \int_1^{\infty} \frac{1}{x^2}\,dx \quad \text{converges.}$$

(b) $\displaystyle\int_1^{\infty} \frac{1}{\sqrt{x^2 - 0.1}}\,dx$ diverges because

$$\frac{1}{\sqrt{x^2 - 0.1}} \ge \frac{1}{x} \quad \text{on} \quad [1, \infty) \quad \text{and} \quad \int_1^{\infty} \frac{1}{x}\,dx \quad \text{diverges.}$$

THEOREM —Limit Comparison Test If the positive functions f and g are continuous on $[a, \infty)$, and if

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = L, \qquad 0 < L < \infty,$$

then

$$\int_a^{\infty} f(x)\,dx \qquad \text{and} \qquad \int_a^{\infty} g(x)\,dx$$

both converge or both diverge.

EXAMPLE Show that

$$\int_1^{\infty} \frac{dx}{1 + x^2}$$

converges by comparison with $\int_1^{\infty} (1/x^2)\, dx$. Find and compare the two integral values.

Solution The functions $f(x) = 1/x^2$ and $g(x) = 1/(1 + x^2)$ are positive and continuous on $[1, \infty)$. Also,

$$\begin{aligned} \lim_{x \to \infty} \frac{f(x)}{g(x)} &= \lim_{x \to \infty} \frac{1/x^2}{1/(1 + x^2)} = \lim_{x \to \infty} \frac{1 + x^2}{x^2} \\ &= \lim_{x \to \infty} \left(\frac{1}{x^2} + 1 \right) = 0 + 1 = 1, \end{aligned}$$

a positive finite limit Therefore, $\int_1^{\infty} \frac{dx}{1 + x^2}$ converges because $\int_1^{\infty} \frac{dx}{x^2}$ converges.

The integrals converge to different values, however:

$$\int_1^{\infty} \frac{dx}{x^2} = \frac{1}{2 - 1} = 1$$

and

$$\begin{aligned} \int_1^{\infty} \frac{dx}{1 + x^2} &= \lim_{b \to \infty} \int_1^{b} \frac{dx}{1 + x^2} \\ &= \lim_{b \to \infty} [\tan^{-1} b - \tan^{-1} 1] = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}. \end{aligned}$$

EXAMPLE Investigate the convergence of $\displaystyle\int_1^\infty \frac{1-e^{-x}}{x}\,dx.$

Solution The integrand suggests a comparison of $f(x) = (1 - e^{-x})/x$ with $g(x) = 1/x$. However, we cannot use the Direct Comparison Test because $f(x) \le g(x)$ and the integral of $g(x)$ *diverges*. On the other hand, using the Limit Comparison Test we find that

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{x\to\infty} \left(\frac{1-e^{-x}}{x}\right)\left(\frac{x}{1}\right) = \lim_{x\to\infty} (1 - e^{-x}) = 1,$$

which is a positive finite limit. Therefore, $\displaystyle\int_1^\infty \frac{1-e^{-x}}{x}\,dx$ diverges because $\displaystyle\int_1^\infty \frac{dx}{x}$ diverges.

Note that the values

do not appear to approach any fixed limiting value as $b \to \infty$.

EX. Show that $\int_0^\infty e^{-x^2}\,dx$ converges, and find an upper bound for its value.

Solution We can't integrate e^{-x^2}, but we can integrate e^{-x}. We would like to use the inequality $e^{-x^2} \le e^{-x}$, but this is only valid for $x \ge 1$. Therefore, we break the integral into two parts.

On $[0, 1]$ we have $0 < e^{-x^2} \le 1$, so

$$0 < \int_0^1 e^{-x^2}\,dx \le \int_0^1 dx = 1.$$

On $[1, \infty)$ we have $x^2 \ge x$, so $-x^2 \le -x$ and $0 < e^{-x^2} \le e^{-x}$. Thus,

$$0 < \int_1^\infty e^{-x^2}\,dx \le \int_1^\infty e^{-x}\,dx = \lim_{R\to\infty} \left.\frac{e^{-x}}{-1}\right|_1^R$$

$$= \lim_{R\to\infty} \left(\frac{1}{e} - \frac{1}{e^R}\right) = \frac{1}{e}.$$

Hence, $\int_0^\infty e^{-x^2}\,dx$ converges and its value is not greater than $1 + (1/e)$.

EX.

Determine whether $\int_0^\infty \frac{dx}{\sqrt{x + x^3}}$ converges.

The integral is improper of both types, so we write

$$\int_0^\infty \frac{dx}{\sqrt{x+x^3}} = \int_0^1 \frac{dx}{\sqrt{x+x^3}} + \int_1^\infty \frac{dx}{\sqrt{x+x^3}} = I_1 + I_2.$$

On $(0, 1]$ we have $\sqrt{x+x^3} > \sqrt{x}$, so $\qquad I_1 < \int_0^1 \frac{dx}{\sqrt{x}} = 2$

On $[1, \infty)$ we have $\sqrt{x+x^3} > \sqrt{x^3}$, so $\qquad I_2 < \int_1^\infty x^{-3/2}\, dx = 2$

Hence, the given integral converges, and its value is less than 4.

1) Determine whether the following integral is convergent or divergent

$$\int_1^{\infty} \frac{1+\sin\sqrt{x}}{\sqrt{x}+x^3}\,dx = ?$$

<u>Solution:</u> $\int_1^{\infty} \frac{2}{x^3}\,dx$ is convergent by p-test

$$\frac{1+\overset{\le 1}{\sin\sqrt{x}}}{\sqrt{x}+x^3} \le \frac{2}{\sqrt{x}+x^3} \le \frac{2}{x^3},$$ so by comparison

$$\int_1^{\infty} \frac{1+\overset{\le 1}{\sin\sqrt{x}}}{\sqrt{x}+x^3}\,dx$$ is convergent.

) Evaluate the integral $\int_1^{\infty} \frac{\ln x}{x^3}\,dx$, if it is converge

$\frac{\ln x}{x^3} < \frac{x}{x^3} = \frac{1}{x^2}$; $\int_1^{\infty} \frac{1}{x^2}\,dx$ convergent, so

$\int_1^{\infty} \frac{\ln x}{x^3}\,dx$ is convergent.

$$\int_1^{\infty} \frac{\ln x}{x^3}\,dx = \lim_{t\to\infty} \left[\ln x \cdot \left(-\frac{1}{2x^2}\right)\Big|_1^t + \int_1^t \frac{dx}{2x^3} \right]$$

$\ln x = u \Rightarrow \frac{dx}{x} = du$

$\frac{dx}{x^3} = dv \Rightarrow -\frac{1}{2x^2} = v$

$$= \lim_{t\to\infty} \left[\left(\ln t \cdot \left(-\frac{1}{2t^2}\right) - \ln 1 \cdot \left(-\frac{1}{2\cdot 1}\right)\right) - \left(\frac{1}{4t^2}\right) + \frac{1}{4} \right]$$

$$= \frac{1}{4}$$

$$\lim_{t\to\infty} \left(-\frac{\ln t}{2t^2}\right) \to 0$$

... l'Hôpital rule

(3) Is the following integral convergent or divergent?
(Note you are not asked to evaluate the integral)

$$\int_8^\infty \frac{x+1}{\sqrt{x^4-x}}\,dx \,.$$

$$x>0\,; \quad \frac{x+1}{\sqrt{x^4-x}} > \frac{x}{\sqrt{x^4-x}} > \frac{x}{\sqrt{x^4}} = \frac{x}{x^2} = \frac{1}{x}$$

Now, $\int_8^\infty \frac{1}{x}\,dx$ diverges, So by comparison

$$\int_8^\infty \frac{x+1}{\sqrt{x^4-x}}\,dx$$ diverges.

(4) Is the following integral convergent or divergent?

$$\int_0^{\pi/2} \frac{\sin x}{x^{3/2}}\,dx \,.$$

$f(x) = \dfrac{\sin x}{x^{3/2}}$ is not defined and continuous at $x=0$.

for $x \in (0, \pi/2]$, $g(x) = \sin x$

$$\sin x < x \implies 0 < \frac{\sin x}{x^{3/2}} < \frac{x}{x^{3/2}} = \frac{1}{x^{1/2}}$$

$\displaystyle\int_0^{\pi/2} \frac{dx}{x^{1/2}}$ converges, because $p = \frac{1}{2} < 1$.

So by comparison,

$\displaystyle\int_0^{\pi/2} \frac{\sin x}{x^{3/2}}\,dx$ converges.

(5) Homework: Evaluate $\int_0^2 \frac{\ln x}{\sqrt{2x}}\,dx$.

Answer: $(2\ln 2-4)$

(6) Homework : Evaluate $\int_0^{\pi} \frac{dx}{1+\cos x}$.

Answer: (Diverges)

(7) Homework: Evaluate $\int_1^{\infty} \frac{dx}{\sqrt{x^2-0.1}}$.

Answer : (Diverges)

(8) Determine whether the following integral is convergent $\int_1^{\infty} \frac{\sin^2 x}{x^2}\,dx$

(9) $\displaystyle\int_1^\infty \frac{x-1}{x^2+5}\,dx = ?$

$\displaystyle\int_1^\infty \frac{dx}{x}$ diverges $(p=1)$ and

$$\lim_{x\to\infty} \frac{\frac{x-1}{x^2+5}}{\frac{1}{x}} = 1 .$$

Thus,

$\displaystyle\int_1^\infty \frac{x-1}{x^2+5}\,dx$ diverges because $0<1<\infty$ and $\displaystyle\int_1^\infty \frac{dx}{x}$ diverges.

(10) $\displaystyle\int_1^\infty \frac{x^2+7x}{x^5+x^4-2x+5}\,dx = ?$

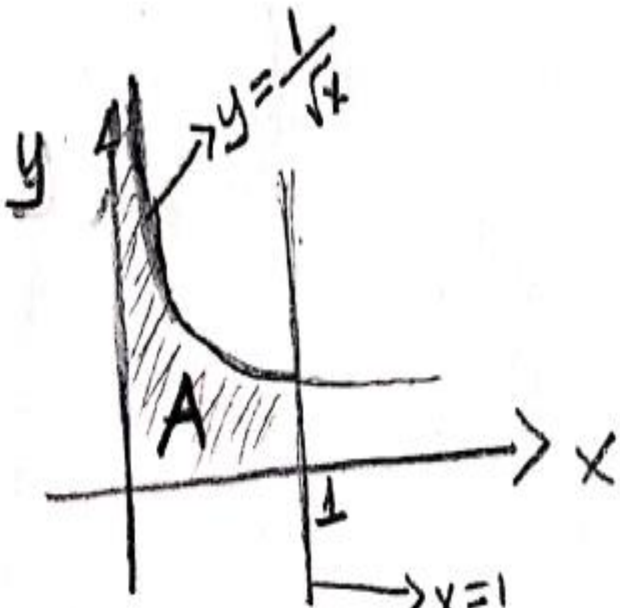
$\displaystyle\int_1^\infty \frac{dx}{x^3}$ converges $(p=3)$.

$$\lim_{x\to\infty} \frac{\frac{x^2+7x}{x^5+x^4-2x+5}}{\frac{1}{x^3}} = 1 .\quad \text{Thus,}$$

$\displaystyle\int_1^\infty \frac{x^2+7x}{x^5+x^4-2x+5}\,dx$ converges because $0<1<\infty$

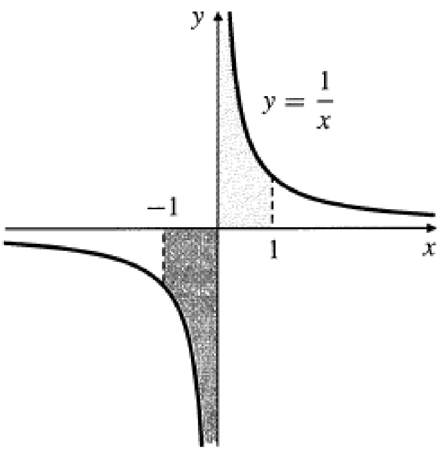
and $\displaystyle\int_1^\infty \frac{dx}{x^3}$ converges.

(11)



Find the shaded area?

Answer: A= 2



EXAMPLE We know that $\frac{d}{dx} \ln|x| = \frac{1}{x}$ if $x \neq 0$. It is *incorrect*, however, to state that

$$\int_{-1}^{1} \frac{dx}{x} = \ln|x| \Big|_{-1}^{1} = 0 - 0 = 0,$$

even though $1/x$ is an odd function. In fact, $1/x$ is undefined and has no limit at $x = 0$, and it is not integrable on $[-1, 0]$ or $[0, 1]$. Observe that

$$\lim_{c \to 0+} \int_{c}^{1} \frac{1}{x} \, dx = \lim_{c \to 0+} -\ln c = \infty,$$

so both shaded regions in Figure have infinite area. Integrals of this type are called **improper integrals**.

EXAMPLE Is the area under the curve $y = (\ln x)/x^2$ from $x = 1$ to $x = \infty$ finite? If so, what is its value?

Solution We find the area under the curve from $x = 1$ to $x = b$ and examine the limit as $b \to \infty$. If the limit is finite, we take it to be the area under the curve. The area from 1 to b is

$$\int_1^b \frac{\ln x}{x^2}\,dx = \left[(\ln x)\left(-\frac{1}{x}\right)\right]_1^b - \int_1^b \left(-\frac{1}{x}\right)\left(\frac{1}{x}\right)dx$$

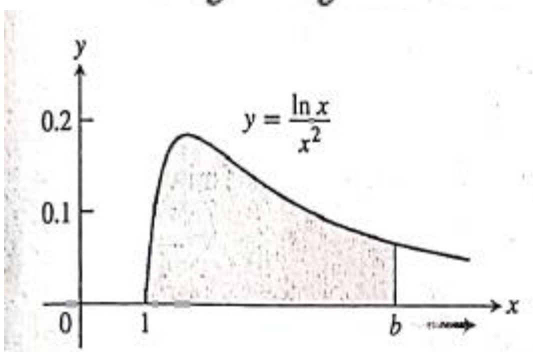
Integration by parts with $u = \ln x$, $dv = dx/x^2$, $du = dx/x$, $v = -1/x$

$$= -\frac{\ln b}{b} - \left[\frac{1}{x}\right]_1^b$$

$$= -\frac{\ln b}{b} - \frac{1}{b} + 1.$$

$\ln x = u \to du = \frac{dx}{x}$

$dv = \frac{dx}{x^2} \to v = -\frac{1}{x}$



The limit of the area as $b \to \infty$ is

$$\int_1^\infty \frac{\ln x}{x^2}\,dx = \lim_{b\to\infty} \int_1^b \frac{\ln x}{x^2}\,dx$$

$$= \lim_{b\to\infty} \left[-\frac{\ln b}{b} - \frac{1}{b} + 1\right]$$

$$= -\left[\lim_{b\to\infty} \frac{\ln b}{b}\right] - 0 + 1$$

$$= -\left[\lim_{b\to\infty} \frac{1/b}{1}\right] + 1 = 0 + 1 = 1. \qquad \text{l'Hôpital's Rule}$$

Thus, the improper integral converges and the area has finite value 1. ∎